NINE EDUCATION IIT ACADEMY

HEAT TRANSFER — PRACTICE ASSIGNMENT

Class 11 Physics · JEE Main Focus · 71 Questions

Conduction · Thermal Resistance · Junctions & Networks · Radiation · Stefan's Law · Wien's Law

FORMULA SHEET The ideas in simple words, then the formula

1. Three ways heat travels

Conduction needs a medium. Particles vibrate and pass energy to their neighbours; the particles themselves do not move along. **Convection** needs a fluid. The hot fluid itself rises and carries the heat. **Radiation** is carried by electromagnetic waves. It needs no medium and crosses vacuum.

2. Heat current (how fast heat flows through a rod)

$$i = \frac{dQ}{dt} = \frac{K A\,\Delta T}{l} \qquad \text{(unit: watt)}$$

More heat flows when the material conducts better (K), the rod is thicker (A) or the ends differ more in temperature (ΔT). Less flows when the rod is longer (l). K = thermal conductivity, unit W m^{-1} K^{-1}. Usually $K_{\text{solid}} > K_{\text{liquid}} > K_{\text{gas}}$.

Steady state: after a long time the temperature at each point stops changing. Every cross-section then carries the same i, and the temperature falls in a straight line along the rod:

$$T(x) = T_H - \frac{(T_H - T_C)}{l}\,x$$

3. Thermal resistance (heat flow behaves like electric current)

$$R = \frac{l}{KA} \qquad \Delta T = i\,R \qquad (\text{like } V = IR)$$

l is the length *along* the heat flow; A is the area *across* it. Unit of R: K/W.

4. Ice melting at one end

Heat that flows in time t melts the ice: $\quad i\,t = m\,L_f \;\Rightarrow\; m = \dfrac{i\,t}{L_f}$

5. Rods in series and in parallel

Series (one after another): the same heat current passes through each rod. Resistances add, and the rod with the bigger R takes the bigger temperature drop.

$$R_{\text{eq}} = R_1 + R_2 + \cdots \qquad K_E = \frac{l_1 + l_2 + \cdots}{\dfrac{l_1}{K_1} + \dfrac{l_2}{K_2} + \cdots} \quad (\text{same } A)$$

Parallel (side by side between the same two temperatures): every rod has the same ΔT, and the heat currents add.

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots \qquad K_E = \frac{K_1A_1 + K_2A_2 + \cdots}{A_1 + A_2 + \cdots} \quad (\text{same } l)$$

6. Junctions

At a junction, heat flowing in = heat flowing out (total heat current = 0, like Kirchhoff's current law). Write $\frac{T_{\text{end}} - T}{R}$ for every rod and add them to zero. Balanced bridge: if $R_1 R_4 = R_2 R_3$, no heat flows through the middle rod.

7. Words used in radiation

Radiation falling on a surface is partly absorbed, reflected and transmitted: $a + r + t = 1$. **Emissive power** E = energy radiated per m^2 per second (W/m^2, depends on T). **Emissivity** $e = E/E_{\text{black}}$ at the same temperature (0 to 1, no unit). **Black body:** absorbs everything, so $a = 1$ and $e = 1$. In practice, a small hole in a hollow cavity. Every body above 0 K radiates all the time. At the same temperature as its surroundings it absorbs as much as it emits.

8. Kirchhoff's law

$$e = a$$

A good absorber is a good emitter; a poor absorber is a poor emitter.

9. Stefan–Boltzmann law (how much power is radiated)

$$P = e\sigma A T^4 \qquad P_{\text{net}} = e\sigma A\left(T^4 - T_0^4\right) \qquad \text{sphere: } P \propto r^2 T^4$$

T must be in kelvin. Doubling the absolute temperature makes the power $2^4 = 16$ times.

10. Wien's law (where the glow is brightest)

$$\lambda_m T = b \qquad \lambda_1 T_1 = \lambda_2 T_2 \qquad \text{with Stefan: } P \propto \frac{r^2}{\lambda_m^4}$$

A hotter body peaks at a shorter wavelength: dull red, then orange, then white.

STANDARD VALUES (used unless a question states otherwise)

QUANTITY	VALUE
Stefan–Boltzmann constant σ	5.67×10^{-8} W m^{-2} K^{-4}
Wien's constant b	2.9×10^{-3} m K
Latent heat of fusion of ice L_f	3.36×10^5 J/kg (80 cal/g)
Kelvin from Celsius	$T(\text{K}) = T(^\circ\text{C}) + 273$

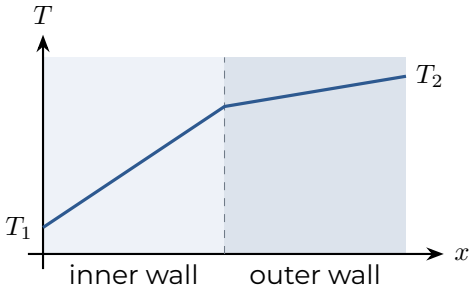
SECTION A — Heat Current & Steady State Q1–9

1. A rod of length 50 cm and cross-section 4 cm^2 has $K = 200$ W m^{-1} K^{-1}. Its ends are held at 100°C and 20°C. Find the heat current.
2. For the rod in Q1, find the heat conducted in 2 minutes.
3. A glass window is 1.5 m $\times$ 1.0 m and 4 mm thick ($K = 0.8$ W m^{-1} K^{-1}). Its inner and outer faces are at 20°C and 5°C. Find the heat lost per second.
4. For the rod in Q1, find the temperature gradient along the rod and the temperature 20 cm from the hot end.
5. A uniform rod 2 m long has its ends at 120°C and 20°C. At what distance from the hot end is the temperature 50°C?
6. Two rods P and Q are made of the same material. P has length l and area A; Q has length $2l$ and area $3A$. Both have the same temperature difference across them. Find $i_P : i_Q$.
7. A rod 1 m long with cross-section 2 cm^2 and $K = 400$ W m^{-1} K^{-1} has one end in boiling water at 100°C and the other in a large block of ice at 0°C. Find the mass of ice that melts in 10 minutes.
8. A thermocol ice box has total wall area 0.8 m^2 and wall thickness 2 cm ($K = 0.04$ W m^{-1} K^{-1}). Ice inside is at 0°C and the room is at 30°C. How much ice melts in 1 hour?
9. Using $i = \dfrac{KA\,\Delta T}{l}$, find the SI unit and the dimensional formula of thermal conductivity K.

JEE MAIN PYQs · Heat Current & Steady State Q10–12

10. A heat source at 10^3 K is connected to another reservoir at 10^2 K by a copper slab 1 m thick ($K = 0.1$ W K^{-1} m^{-1}). Find the energy flux through the slab in the steady state. **[JEE Main 2019, 10 Jan Shift 1]**

11. The wall of a house is made of two different materials of the same thickness. The inner face is at T_1 and the outer face at T_2 ($T_1 < T_2$). In steady state the temperature varies as shown. Which material has the greater thermal conductivity, and why? **[JEE Main 2026, 21 Jan Shift 1]**



12. An ice cube 60 cm × 50 cm × 20 cm is placed in an insulation box of wall thickness 1 cm ($K = 0.05$ W m^{-1} °C^{-1}). The box, keeping the ice at 0°C, is brought into a room at 40°C. Find the rate of melting of ice ($L_f = 3.4 \times 10^5$ J/kg). **[JEE Main 2022, 26 July Shift 2]**

SECTION B — Thermal Resistance, Series & Parallel Q13–25

13. Find the thermal resistance of a copper rod of length 40 cm, cross-section 2 cm^2, $K = 400$ W m^{-1} K^{-1}.

14. A block measures 10 cm × 5 cm × 2 cm and has $K = 50$ W m^{-1} K^{-1}. Find its thermal resistance when heat flows along the 10 cm, the 5 cm and the 2 cm side.

15. Two slabs of the same area 0.1 m^2 are placed face to face. Slab A: thickness 2 cm, $K = 40$ W m^{-1} K^{-1}. Slab B: thickness 3 cm, $K = 20$ W m^{-1} K^{-1}. The outer face of A is at 100°C and of B at 0°C. Find the heat current and the temperature of the interface.

16. Two rods of the same length and area are joined end to end. The first has three times the conductivity of the second. The free ends are held at 100°C (first rod) and 20°C (second rod). Find the junction temperature.

17. Three rods of identical size with conductivities K, $2K$ and $4K$ are joined in that order. The free ends are at 140°C and 0°C. Find the temperatures of the two joints.

18. A wall of area 10 m^2 has 20 cm of brick ($K = 0.6$ W m^{-1} K^{-1}) and 5 cm of insulation ($K = 0.04$ W m^{-1} K^{-1}). Outside is at 40°C and inside at 25°C. Find the heat entering the room per second.

19. Two rods of the same length l and area A, with conductivities K and $3K$, are placed side by side between 80°C and 20°C. Find the total heat current and the effective conductivity of the pair.

20. Two slabs of equal thickness with conductivities $2K$ and $3K$ are joined face to face. Find the effective conductivity.

21. Two rods of the same area are joined end to end: 1 cm with $K = 100$ W m^{-1} K^{-1} and 3 cm with $K = 50$ W m^{-1} K^{-1}. Find the effective conductivity.

22. Two rods of the same length are placed side by side: $K_1 = 100\,\text{W m}^{-1}\,\text{K}^{-1}$, $A_1 = 2\,\text{cm}^2$; $K_2 = 40\,\text{W m}^{-1}\,\text{K}^{-1}$, $A_2 = 3\,\text{cm}^2$. Find the effective conductivity.

23. A copper rod ($K = 400\,\text{W m}^{-1}\,\text{K}^{-1}$) and a steel rod ($K = 50\,\text{W m}^{-1}\,\text{K}^{-1}$), each 50 cm long with cross-section 1 cm^2, are joined end to end. The free copper end is at 100°C and the free steel end at 0°C. Find the heat current and the junction temperature.

24. A rod connects two reservoirs. It is cut into two equal halves, and the halves are placed side by side between the same reservoirs. By what factor does the heat current change?

25. Two identical rods joined end to end melt a certain mass of ice in 12 minutes. If the same rods are placed side by side between the same reservoirs, how long will the same mass of ice take to melt?

JEE MAIN PYQs · Thermal Resistance, Series & Parallel Q26–34

26. Two plates A and B have conductivities 84 W m^{-1} K^{-1} and 126 W m^{-1} K^{-1}, with the same area and thickness. They are placed in contact, and their outer surfaces are kept at 100°C and 0°C. Find the temperature of the contact surface. **[JEE Main 2023, 13 Apr Shift 2]**

27. Two materials of conductivity $3K$ and K and thickness d and $3d$ are joined to form a slab. The outer surface of the $3K$ layer is at θ_2 and of the K layer at θ_1 ($\theta_2 > \theta_1$). Find the interface temperature. **[JEE Main 2019, 09 Apr Shift 2]**

28. Two cylindrical rods A and B are joined in a straight line with $L_A/L_B = 1/2$, $r_A/r_B = 2$ and $K_A/K_B = 1/2$. The free ends of A and B are at 400 K and 200 K. Find the interface temperature. **[JEE Main 2025, 07 Apr Shift 2]**

29. A steel rod (1) and a copper rod (2) are joined end to end with $K_2/K_1 = 9$, $A_1/A_2 = 2$ and $L_1/L_2 = 2$. The free steel end is at 450°C and the free copper end at 0°C. Find the junction temperature. **[JEE Main 2022]**

30. Blocks M_1 (16 cm long) and M_2 (8 cm long) of the same cross-section are joined. The free end of M_1 is at 100°C, the junction at 80°C and the free end of M_2 at 0°C. If M_2 has conductivity K, find the conductivity of M_1. **[JEE Main 2022, 24 June Shift 1]**

31. Three rods of identical size with conductivities K_1, K_2, K_3 are joined end to end. One end is at 100°C and the other at 0°C; the joints are at 70°C and 20°C. Find $K_1 : K_2 : K_3$. **[JEE Main 2020, 06 Sep Shift 2]**

32. Two identical wires of conductivities K_1 and K_2 are joined in series. Find the effective conductivity. **[JEE Main 2021, 17 Mar Shift 1]**

33. Plates A (K, 4.0 cm thick) and B ($2K$, 2.5 cm thick), each of area 120 cm^2, form a compound plate. Its equivalent conductivity is $\left(1 + \frac{5}{\alpha}\right) K$. Find α. **[JEE Main 2022, 29 June Shift 1]**

34. A cylinder of radius R (conductivity K_1) is surrounded by a cylindrical shell of inner radius R and outer radius $2R$ (conductivity K_2). Find the effective conductivity for heat flowing along the length. **[JEE Main 2019, 12 Jan Shift 1]**

SECTION C — Junctions & Networks Q35–41

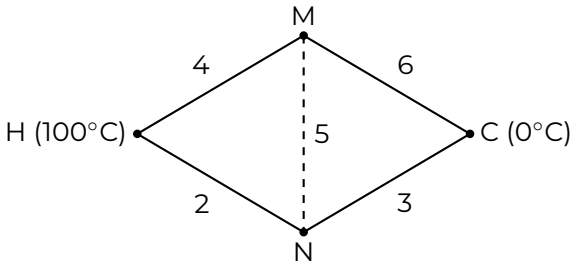
35. Three identical rods meet at a junction. Their free ends are at 0°C, 30°C and 90°C. Find the junction temperature.

36. Three rods of identical size with conductivities K, $2K$ and $3K$ meet at a junction. Their free ends are at 0°C, 50°C and 100°C respectively. Find the junction temperature.

37. Three identical rods meet at a junction. The free end of one is at 100°C; the free ends of the other two are at 0°C. Find the junction temperature, and compare the heat current in the hot rod with that in each cold rod.

38. A uniform rod AB of length $3L$ has A at 90°C and B at 0°C. Points C and D on the rod are at distance L from A and L from B. A second rod of the same material and cross-section, length L, is joined between C and D. Find the temperatures of C and D.

39. Five rods are joined as shown, with thermal resistances in K/W. Point H is at 100°C and point C at 0°C. Is the bridge balanced? Find the heat current through the middle rod, the total heat current, and the temperatures of M and N.



40. Four identical rods form a square ABCD. Corner A is held at 100°C and the opposite corner C at 0°C. Find the temperatures of B and D. If a fifth rod is joined along the diagonal BD, how much heat flows through it?

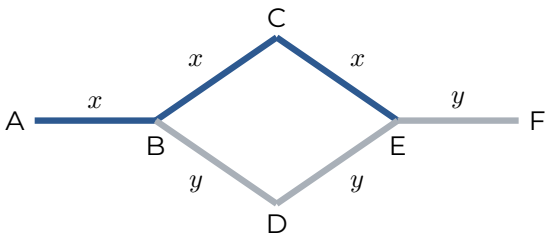
41. Three identical rods form a triangle ABC. A is held at 100°C and B at 0°C. Find the temperature of C and the ratio of the heat current along AB to that along the path ACB.

JEE MAIN PYQs · Junctions & Networks Q42–44

42. Three conductors of the same length have $k_1 = 60$, $k_2 = 120$ and $k_3 = 135$ J s^{-1} m^{-1} K^{-1}. Conductors 1 and 2 have the same area A and are placed side by side between 100°C and a junction at θ. Conductor 3 has area $2A$ and runs from the junction to 0°C. Find θ. **[JEE Main 2025, 22 Jan Shift 1]**

43. A temperature difference of 120°C is maintained across a uniform rod AB of length $2L$. A bent rod PQ of the same cross-section and length $3L/2$ is joined across AB at points P and Q, each $L/2$ from the nearer end. Find the temperature difference between P and Q in steady state. **[JEE Main 2019, 09 Jan Shift 1]**

44. Rods x and y of equal size but different materials are joined as shown. A is at 100°C and F at 40°C. The conductivity of x is three times that of y. Find the temperatures of junctions B and E. **[JEE Main 2026, 22 Jan Shift 1]**



SECTION D — Radiation Basics & Kirchhoff's Law Q45–51

45. 500 J of radiation falls on a surface. 150 J is reflected and 50 J passes through. Find a, r and t.

46. An opaque surface absorbs 70% of the radiation falling on it. Find its reflectance and its emissivity.

47. True or false: (a) The emissive power of a perfectly black body is 1. (b) Every body above 0 K emits radiation. (c) A body at the same temperature as its surroundings stops radiating. (d) The emissivity of a body is always less than its absorptive power.

48. Why does a small hole in a hollow box behave as a black body, while the box itself does not?

49. A white plate has a black spot painted on it. It is heated strongly and taken into a dark room. Which part glows brighter, and why?

50. A piece of red glass is heated until it glows. In which colour does it glow most strongly? Explain using Kirchhoff's law.

51. A surface has emissive power 4000 W/m^2. A black body at the same temperature has emissive power 10000 W/m^2. Find the emissivity and the absorptive power of the surface.

JEE MAIN PYQs · Radiation Basics & Kirchhoff's Law Q52

52. How many of these statements about a black body are *incorrect*? (A) It emits or absorbs energy as electromagnetic radiation. (B) The frequency distribution of its radiation depends on temperature. (C) At a given temperature, the intensity vs frequency curve passes through a maximum. (D) The maximum of the curve is at a higher frequency at a higher temperature. **[JEE Main 2023, April Shift 1 (Chemistry paper)]**

SECTION E — Stefan–Boltzmann Law Q53–61

53. A black body is heated from 27°C to 327°C. By what factor does its radiated power increase?

54. The absolute temperature of a black body is increased by 50%. Find the percentage increase in the power it radiates.

55. A black sphere of radius 10 cm is at 727°C. Find the power it radiates (ignore the surroundings).

56. Two spheres of the same material have radii r and $2r$ and temperatures $2T$ and T. Find the ratio of the powers they radiate.

57. A black body of surface area 0.5 m^2 is at 400 K in surroundings at 300 K. Find its net rate of heat loss.

58. A body of surface area 0.02 m^2 at 1000 K radiates 567 W. Find its emissivity (ignore the surroundings).

59. A metal sphere inside an evacuated chamber at 300 K is held at 600 K by a heater using 150 W. When the sphere is blackened, the heater needs 500 W to hold the same temperature. Find the emissivity of the metal.

60. Two black spheres radiate the same power. The first is at twice the absolute temperature of the second. Find the ratio of their radii r_1/r_2.

61. The Sun (radius 7×10^8 m) is a black body at 5800 K, at a distance of 1.5×10^{11} m from the Earth. Estimate the power received per m^2 at the Earth (the solar constant).

JEE MAIN PYQs · Stefan–Boltzmann Law Q62–64

62. Two spheres of the same material have radii 0.2 m and 0.8 m and are at 800 K and 400 K. If the smaller one radiates energy E, how much does the bigger one radiate? (Neglect the surroundings.) **[JEE Main 2025, 22 Jan Shift 1]**

63. A bulb filament wire is 10 cm long with diameter 0.5 mm. It is at 1727°C and radiates 94.2 W. Its emissivity is $x/8$. Find x. (Take $\sigma = 6.0 \times 10^{-8}$ W m^{-2} K^{-4}, $\pi = 3.14$.) **[JEE Main 2025, 07 Apr Shift 1]**

64. Parallel rays of intensity 912 W/m^2 fall on a black sphere in surroundings at 300 K. Taking $\sigma = 5.7 \times 10^{-8}$ W m^{-2} K^{-4}, find the steady-state temperature of the sphere. **[JEE Main 2014, older, extra practice]**

SECTION F — Wien's Law Q65–70

65. The Sun's radiation peaks at 580 nm. Estimate its surface temperature.

66. Find the wavelength at which a human body at 37°C radiates most strongly. In which region of the spectrum is it?

67. A black body at 2000 K peaks at wavelength λ. Find the peak wavelength when it is heated to 3000 K.

68. The peak wavelength of a black body shifts from 2.0 μm to 1.5 μm. Find the ratio of the new to the old temperature, and the factor by which its radiated power changes.

69. Two black discs of the same radius have peak wavelengths 400 nm and 800 nm. Find the ratio of the powers they radiate.

70. Three black spheres have radii 1 cm, 2 cm and 3 cm and peak wavelengths 300 nm, 400 nm and 500 nm. Which radiates the most power?

JEE MAIN PYQs · Wien's Law Q71

71. Two spherical stars A and B emit black body radiation. The radius of A is 400 times that of B, and A emits 10^4 times the power of B. Find the ratio λ_A/λ_B of their peak wavelengths. **[JEE Advanced 2015, extra practice]**

ANSWER KEY Q1–71

Q#	ANSWER	Q#	ANSWER
1	12.8 W	37	≈ 33.3°C; hot rod carries $2\times$ each cold rod
2	1536 J	38	$T_C = 54$°C, $T_D = 36$°C
3	4500 W	39	Balanced; 0 W; 30 W; M, N both 60°C
4	160°C/m; 68°C	40	B, D both 50°C; zero along BD
5	1.4 m	41	$T_C = 50$°C; 2 : 1
6	2 : 3	42	40°C
7	≈ 14.3 g	43	45°C
8	≈ 514 g	44	≈ 89°C and ≈ 73°C*
9	W m^{-1} K^{-1}; [M L T^{-3} K^{-1}]	45	$a = 0.6$, $r = 0.3$, $t = 0.1$
10	90 W/m^2	46	$r = 0.3$, $e = 0.7$
11	Outer wall (smaller slope, larger K)	47	(a) F (b) T (c) F (d) F
12	$\approx 61 \times 10^{-5}$ kg/s	48	Light inside reflects many times and is absorbed
13	5 K/W	49	Black spot (good absorber = good emitter)
14	2 K/W, 0.5 K/W, 0.08 K/W	50	Greenish (absorbs green, so emits green)
15	5000 W; 75°C	51	$e = a = 0.4$
16	80°C	52	0 (all four are correct)
17	60°C and 20°C	53	16 times
18	≈ 95 W	54	406.25%
19	$240KA/l$; $K_E = 2K$	55	≈ 7.1 kW
20	$2.4K$	56	4 : 1
21	$400/7 \approx 57.1$ W m^{-1} K^{-1}	57	≈ 496 W
22	64 W m^{-1} K^{-1}	58	0.5
23	≈ 0.89 W; ≈ 88.9°C	59	0.3
24	Becomes 4 times	60	$1/4$
25	3 minutes	61	≈ 1.4 kW/m^2
26	40°C	62	E
27	$\frac{\theta_1}{10} + \frac{9\theta_2}{10}$	63	5
28	360 K	64	≈ 330 K
29	45°C	65	5000 K
30	$8K$	66	≈ 9.4 μm, infrared
31	10 : 6 : 15	67	$2\lambda/3$
32	$\frac{2K_1K_2}{K_1+K_2}$	68	$4/3$; power $\times\, 256/81 \approx 3.16$
33	21	69	16 : 1
34	$\frac{K_1+3K_2}{4}$	70	The 2 cm sphere (400 nm)
35	40°C	71	2
36	$200/3 \approx 66.7$°C		

*Q44: some published keys print 80°C and 70°C. Working the network gives 89°C and 73°C (B to E is $R_y/2$, total $11R_y/6$).